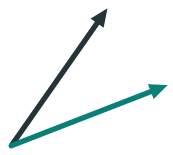
01

Bring them together.

They match.

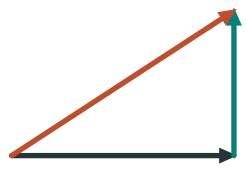


Turn one. Compare again.

02

Take this step.

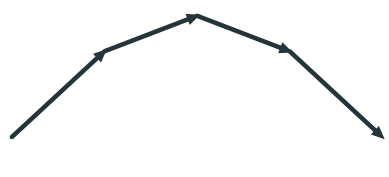
Then take this step.



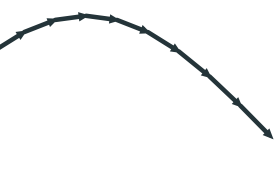
One step reaches the same end.

03

A few steps.



More, smaller steps.

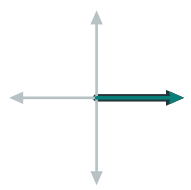


Follow the change more closely.

04

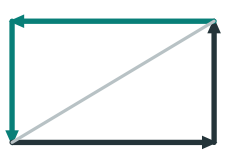
Start together.

Turn just one.

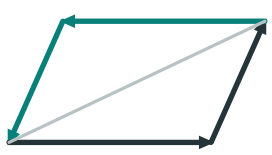


Keep turning the same way.
It comes back.

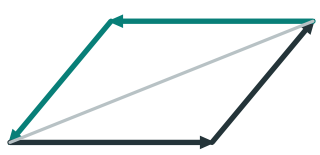
05



Two pieces make this patch.

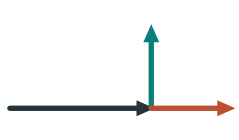


Slide the top sideways.

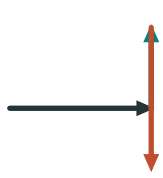


Same area. Different shape.

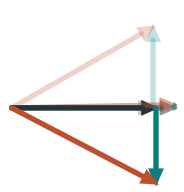
06



Now, black changes
at green’s pace.



At that same instant, green changes
at black’s pace, the opposite way.

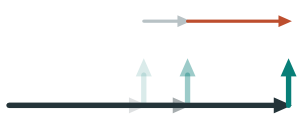


Under both rules,
the joined arrow turns.

07

New rule:
green sets the slide.

Shift, then slide.



Slide, then shift.
Same end.

08

Black advances.

Green stays the same.

Every step keeps that amount.

The mathematics and the debt

Companion to strips 01–08 · supplied structure, derived consequences, explicit limits

What these pictures can establish

This is a supplied classical model with one degree of freedom. It makes a local Hamiltonian construction and one Noether example visible. It does not derive mechanics from marks alone, nor yet explain quantum mechanics.

01–04 · Comparing, composing, refining, returning

We supply an affine plane, comparison by translation, and reversible rotations. Tip-to-tail composition gives vector addition. Strip 3 samples a prescribed differentiable path; drawings alone do not prove a smooth limit. Strip 4 introduces cyclic position (angular phase), not the specialized geometric phase of transport/holonomy, and not a quantum amplitude. It shows three intermediate directions and a return.

05 · The extra structure that makes Hamilton possible

We now supply a STATE plane with dimensionless conjugate coordinates q,p and oriented area $\omega = dq \wedge dp$. In strips 6–8 black measures q (right positive), green measures p (up positive); left/down are negative. The green tip locates the state relative to the black tail. These are state components, not spatial directions. Conjugacy is assumed, not established by a right angle. In physical units p dq has units of action. Momentum need not equal velocity.

05 → 06 · The local derivation

Let u,v be the two rates of change: a dot means change per unit time. Assume an autonomous C^1 rule and preservation of oriented area for EVERY small cell. Local expansion in one component must cancel that in the other: $\partial u/\partial q + \partial v/\partial p = 0$. A partial derivative is a slope while holding the other coordinate fixed. This cancellation lets one function H generate both rates by its two slopes. Formally, $\alpha = u\,dp - v\,dq$ is closed; on a simply connected local patch it equals dH. Thus

$$\dot{q} = u = \partial H/\partial p, \qquad \dot{p} = v = -\partial H/\partial q.$$

These are Hamilton’s equations. H is fixed up to an additive constant. A single patch in strip 5 illustrates the hypothesis, not its general proof. Topology can obstruct a global H; identifying H with physical energy needs a physical model.

06 · An exact turning rule

Choose $H = (q^2+p^2)/2$. The two rate panels use the SAME state (2.1,1.2), at the SAME instant. Orange lengths are 1.2 and 2.1 on a common rate scale. They are not successive kicks. The last panel joins each pair as in strip 2: three exact states at times 0, 0.5, 1 share one origin and scale; pale means earlier. The exact evolution is

$$q' = q\cos h + p\sin h, \qquad p' = p\cos h - q\sin h.$$

This preserves ω and H. The Euler assignment $(q + h\,p, p - h\,q)$ multiplies area and q^2+p^2 by $1+h^2$ instead. Circles belong to this chosen example and scaling, not every Hamiltonian system.

07 · A different rule and its continuous symmetry

Switch to $H = p^2/2$: $\Phi_t(q,p) = (q+t\,p, p)$. A configuration shift $T_a(q,p) = (q+a,p)$ is canonical and leaves H unchanged, so $\Phi_t T_a = T_a \Phi_t$. The two order panels both start at (2,0.65), use shift a=1.5 and time t=1, and finish at (4.15,0.65). Their pale intermediate q values are 3.5 and 2.65. This is a different rule: the oscillator does NOT have this translation symmetry.

08 · The Noether charge and why it is conserved

Translation is generated by $Y = \partial/\partial q$; its pairing with ω gives dp, so the charge is $J = p$. Since H has no q dependence, $dJ/dt = \dot{p} = -\partial H/\partial q = 0$. Equivalently, use $S = \int (p\,\dot{q} - H)\,dt$. Independent interior variations of q,p, with $\delta q = 0$ at time endpoints, give Hamilton’s equations. Constant q translations preserve the action and map endpoint data to translated data; they are not fixed-endpoint variations. Noether conservation holds along stationary trajectories. Equal drawn marks alone do not prove this theorem.

Limits and the remaining debt

Supplied: geometry, oriented pairing, regularity, the two model rules and variational hypotheses. Derived conditionally: local Hamiltonian form, exact example flows and conserved translation charge. Not derived: the earlier Genesis/L1 result, an identity of holonomy with Noether, quantization, Born weights, Everett branching or Bell correlations. Novice comprehension remains to be tested.